

# RFC: Statistical Variant Significance Thresholds

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**Affects:** speconsense (core), speconsense-summarize

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## 1. Problem

Speconsense performs recursive MSA-based variant phasing: it examines the partial order alignment within each cluster, identifies positions where reads disagree, and splits the cluster into haplotypes. The split decision currently uses two fixed thresholds applied locally within each subcluster:

- `--min-variant-frequency 0.10` (at least 10% of reads at that position)
- `--min-variant-count 5` (at least 5 reads supporting the alternative allele)

These thresholds are applied to the *current subcluster size*, not the original specimen read count. During recursive phasing, subclusters shrink at each level. A subcluster of 50 reads (split from an original 1000) can itself split when 5 reads (10%) show an alternative base — even though 5 out of 1000 total reads is well within the expected range of sequencing error.

This creates a risk of **error-driven splits**: the tool reports sequence variants that are statistically indistinguishable from nanopore substitution errors. These false variants consume analyst time during validation and, in automated pipelines, can inflate diversity estimates.

## 2. Solution

We add a statistical significance floor based on the **critical error rate** ( $p^*$ ) framework described in the companion paper ([docs/variant\\_significance/](#)). Before accepting a variant split, speconsense now asks:

“What per-position error rate would make this observation plausible as artifact?”

If the answer is below the assumed platform error rate, the variant is suppressed.

### How it works

At each candidate split position, the tool computes  $p^*$  by solving:

$$L * P(\text{Binom}(N, p^*/3) \geq M) = \alpha$$

where:

| Symbol                    | Meaning                                               | Source                                                       |
|---------------------------|-------------------------------------------------------|--------------------------------------------------------------|
| <b>N</b>                  | Total specimen input reads                            | Fixed at specimen level (not subcluster)                     |
| <b>M</b>                  | Reads supporting the minority allele at this position | From MSA at current recursion level                          |
| <b>L</b>                  | Consensus sequence length                             | From SPOA result at current recursion level                  |
| <b>alpha</b>              | Significance level                                    | User parameter (default 1e-5)                                |
| <b><math>p^*/3</math></b> | Per-base error probability under uniform model        | Assumes errors distribute equally across 3 alternative bases |

The variant is **significant** (allowed to split) when  $p^* \geq$  the assumed error rate. In other words: explaining this many reads as error would require an error rate at least as high as what we assume the platform actually produces. If  $p^*$  is below the assumed rate, the pattern is consistent with normal sequencing error and the split is suppressed.

## Key design choices

**N is total specimen reads, not subcluster size.** This is the conservative choice. A variant that appears in 5 out of 50 reads in a subcluster is really 5 out of the original 1000 specimen reads — the other 950 reads were assigned to other haplotypes at earlier recursion levels. Using the full N prevents deep recursion from manufacturing significance through denominator shrinkage.

**Uniform error model ( $q = p/3$ ).** We assume substitution errors distribute equally across the three alternative bases. This is less conservative than the worst-case model ( $q = p$ , all errors produce the same base) but more realistic for nanopore chemistry. The uniform model yields  $p^*$  values roughly 3x higher than the conservative model, providing comfortable margins for well-supported variants while still catching weak ones.

**Per-variant significance, no cross-variant correction.** Each candidate split position is tested independently. We do not apply additional Bonferroni correction across the set of candidate positions within a cluster, because the positions are not independent (they share the same reads) and the existing frequency/count filters already reduce the candidate set.

**Bonferroni over positions (L).** The correction factor L accounts for multiple testing across all amplicon positions. This is conservative — most positions will not have variant signal — but avoids assumptions about which positions are “testable.”

## 3. User interface

### speconsense (core tool)

Two new CLI parameters in the Variant Phasing group:

```
--assumed-error-rate FLOAT    Assumed per-position error rate for variant
                               significance (uniform model). Set to 0 to disable.
                               (default: 0.015)

--significance-level FLOAT     Significance level (alpha) for variant significance
                               testing. (default: 1e-5)
```

Both are also available as profile keys (`assumed-error-rate`, `significance-level`).

### speconsense-summarize

Two new CLI parameters in the Filtering group:

```
--assumed-error-rate FLOAT    Assumed per-position error rate. Secondary variants
                               with critical error rate (cer) below this value are
                               filtered as potentially artifactual. Primary variants
                               are never filtered. Set to 0 to disable.
                               (default: 0.015)

--no-cer-filter                Disable all CER-based filtering, even when
                               --assumed-error-rate is set.
```

CER filtering in `speconsense-summarize` is applied **after HAC grouping**, only to **secondary variants** within each group. The primary (largest) variant in each group is never filtered by CER — the artifact hypothesis requires a stronger competing signal, and the primary has none. This prevents eliminating single-cluster specimens or dominant variants where the CER question (“artifact of what?”) is not coherent. Variants without a `cer` header (e.g., from older `speconsense` output) are never filtered.

Both are also available as profile keys (`assumed-error-rate`, `no-cer-filter`). The `assumed-error-rate` key can be set once in a profile to apply to both tools.

## FASTA header annotation

When variant phasing produces a split, the output FASTA headers include two new fields:

```
>sample-c1 size=200 ric=100 rid=95.2 rid_min=90.1 cer=0.17 cer.a=1e-05
```

| Field       | Meaning                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| cer=0.17    | Critical error rate ( $p^*$ ) — the per-position error rate that would make this cluster’s read count plausible as artifact. Higher is better. |
| cer.a=1e-05 | The alpha (significance level) used to compute this $p^*$ .                                                                                    |

The **cer** value is computed at output time from the final cluster’s read count (M), total specimen reads (N), and consensus length (L). The annotation answers a simple question: “could this many reads be sequencing error?”

CER is only reported when the value is physically meaningful — specifically, when  $p^* < 0.75$ . Under the uniform error model,  $p^* = 0.75$  is the **signal destruction threshold**: each of the 4 bases becomes equally likely (25% each), and the implied reference population (reads carrying the true sequence) equals the variant count. Above this, the “true sequence” has fewer supporting reads than the variant, making the artifact hypothesis incoherent. In practice, this means dominant clusters (typically >33% of reads at N=1000) receive no **cer** annotation — they are the presumed true sequence, not artifact candidates.

When `--assumed-error-rate 0` is set (CER disabled), no **cer** fields appear.

## FASTA field presets (speconsense-summarize)

The **cer** and **cer\_alpha** fields are available in the `--fasta-fields` system:

- Added to the **qc** preset: **size**, **ric**, **length**, **rid**, **ambig**, **cer**, **cer\_alpha**
- Added to the **full** preset: **size**, **ric**, **length**, **rawric**, **rawlen**, **snp**, **ambig**, **rid**, **cer**, **cer\_alpha**, **primers**
- Available individually: `--fasta-fields default,cer,cer_alpha`

## 4. Defaults and behavior

**Default: 1.5% assumed error rate, alpha = 1e-5**

At the standard operating point (N=1000 presample, L~700 for ITS), this requires roughly **M ≥ 23 reads** for a variant to pass the significance test at the default 1.5% assumed error rate. At lower N, the threshold is a larger fraction of reads (M=9 at N=100, M=6 at N=20), ensuring CER is conservative for small specimens. (See the Appendix for the full matrix.)

For comparison, the existing `--min-variant-count 5` threshold allows splits with as few as 5 reads. The CER gate will suppress most of these at high N, while still allowing them at low N where 5 reads represents stronger evidence.

The default of 1.5% is slightly below the typical ONT R10 per-position substitution rate of ~2%, providing a margin that avoids filtering variants near the boundary while still catching clear artifacts.

### Interaction with existing thresholds

The CER test is applied **after** the existing frequency/count filters (cheap) and **before** the within-cluster error calculation (expensive). A variant must pass all three gates:

1. **Frequency gate**: Alternative allele frequency  $\geq$  `--min-variant-frequency`
2. **Count gate**: Alternative allele count  $\geq$  `--min-variant-count`
3. **CER gate**:  $p^* \geq$  `--assumed-error-rate` (new)

The existing thresholds remain useful as fast pre-filters. The CER gate adds the statistical dimension that accounts for total specimen size.

### Disabling the feature

Setting `--assumed-error-rate 0` disables the CER gate entirely. All variant splits that pass the existing frequency/count thresholds will proceed as before. This is the recommended setting for users who want to preserve exact backward compatibility.

## 5. Practical implications for validators

### What changes in output?

With the default 1.5% assumed error rate:

- **Well-supported variants ( $M \gg 20$ ):** No change. These variants have  $p^*$  well above any realistic error rate and will continue to be reported.
- **Marginal variants ( $M \sim 5-20$  from large specimens):** Some will be suppressed. These are the variants most likely to be error-driven splits. Validators currently spending time on these can expect fewer of them.
- **Small specimens ( $N < 100$ ):** Minimal change. When  $N$  is small, even  $M=5$  represents a substantial fraction, and  $p^*$  will typically exceed the assumed error rate.

### New header fields for triage

The `cer` field provides a direct quality signal for variant triage:

- **`cer > 0.10`:** Very robust. Would require  $>10\%$  per-position error to explain as artifact.
- **`cer = 0.02-0.10`:** Moderate. Significant at the default threshold, but closer to the boundary.
- **`cer < 0.015`:** Below default threshold (would be suppressed by CER gate). If seen, it means the variant was produced with CER disabled or a lower assumed error rate.
- **No `cer` field:** The cluster is too large relative to total specimen reads for the artifact hypothesis to be meaningful ( $p^* \geq 0.75$ ). This is expected for dominant clusters and is not a quality concern.

The `cer` value depends on the alpha used (recorded in the `cer.a` field), so values computed at different significance levels are not directly comparable. However, unlike a p-value, `cer` has a concrete physical interpretation: it is the per-position error rate needed to explain the observation as artifact. As platform chemistry improves and error rates drop, the same `cer` values become more convincing without changing any parameters.

### Comparison with existing quality signals

| Signal           | What it measures                         | Limitation                                           |
|------------------|------------------------------------------|------------------------------------------------------|
| <code>ric</code> | Number of reads used for consensus       | Doesn't account for total specimen size              |
| <code>rid</code> | Mean read-to-consensus identity          | Measures cluster homogeneity, not variant evidence   |
| <code>cer</code> | Statistical strength of variant evidence | Requires<br><code>--assumed-error-rate &gt; 0</code> |

## 6. Examples

### Example 1: Strong variant passes

Specimen with  $N=1000$  reads. Phasing detects 80 reads supporting an alternative allele at a position in a 700bp amplicon.

```
p* = compute_critical_error_rate(N=1000, M=80, L=700, alpha=1e-5)
# p* 0.13 (13%)
```

Since 13% » 1.5% (assumed error rate), this variant passes. Header: `cer=0.13 cer.a=1e-05`.

### Example 2: Weak variant suppressed

Same specimen, N=1000. Deep recursion produces a subcluster where 6 reads show an alternative allele.

```
p* = compute_critical_error_rate(N=1000, M=6, L=700, alpha=1e-5)
# p* 0.003 (0.3%)
```

Since 0.3% < 1.5% (assumed error rate), this variant is suppressed. The split does not occur, and those 6 reads remain in their parent cluster.

### Example 3: Small specimen, same count passes

Specimen with N=20 reads. 6 reads support an alternative allele.

```
p* = compute_critical_error_rate(N=20, M=6, L=700, alpha=1e-5)
# p* 0.07 (7%)
```

Since 7% > 1.5%, this variant passes despite having the same M=6 as Example 2. At N=20, 6 reads represent 30% of the specimen, which is much stronger evidence than 6/1000.

## 7. Migration and backward compatibility

- **Output format:** The `cer=` and `cer.a=` header fields are additive. Downstream tools that don't recognize them will ignore them (they appear after existing fields in the space-delimited header).
- **Default behavior change:** The default `--assumed-error-rate 0.015` means the CER gate is active by default in both tools. Some variants that were previously reported will now be suppressed. To preserve exact previous behavior, set `--assumed-error-rate 0`.
- **speconsense-summarize:** CER filtering is now active by default (`--assumed-error-rate 0.015`), applied after HAC grouping to secondary variants only — primary variants are always retained. Use `--assumed-error-rate 0` or `--no-cer-filter` to disable.
- **Profiles:** The `assumed-error-rate` key works in both `speconsense` and `speconsense-summarize` sections of a profile, allowing a single profile to configure both tools consistently.
- **ConsensusInfo type:** Two new optional fields (`cer`, `cer_alpha`) with `None` defaults. Existing code using `ConsensusInfo` will not break.

## 8. Real-world validation (ONT98 dataset)

The CER framework was validated on the ONT98 dataset (960 wells across 10 ONT runs, ITS amplicons). Of these, 955 produced clustering output and 837 survived summarize filtering (min-RiC). Both `speconsense` and `speconsense-summarize` were run with the default settings (`--assumed-error-rate 0.015`, default `--min-cluster-ratio`).

### Two-stage impact

CER operates at two stages, and the relative contribution of each depends on specimen read count:

1. **Split prevention** (`speconsense` core): The CER gate suppresses weak splits during recursive phasing. Reads that would have formed a separate cluster instead remain in the parent cluster.
2. **Summarize filtering** (`speconsense-summarize`): Post-HAC CER filtering removes secondary variants whose CER falls below the assumed error rate. Primary variants are never filtered.

| Stage               | Variants Removed | Share |
|---------------------|------------------|-------|
| Split prevention    | 323              | 27%   |
| Summarize filtering | 871              | 73%   |
| <b>Total</b>        | <b>1,194</b>     |       |

### Overall impact

| Metric                            | Baseline  | CER 0.015 | Delta           |
|-----------------------------------|-----------|-----------|-----------------|
| Raw clusters (pre-summarize)      | 3,369     | 3,046     | -323 (-9.6%)    |
| Summary variants                  | 2,831     | 1,637     | -1,194 (-42.2%) |
| Merged (multi-component) variants | 497       | 308       | -189            |
| Single-variant specimens          | 246 (29%) | 446 (53%) | +200            |
| HAC groups                        | 1,146     | 1,146     | 0               |
| Specimens                         | 837       | 837       | 0               |

Zero specimens and zero HAC groups were lost. The 1,194 removed variants were overwhelmingly small secondary clusters (RiC 5-12) from high-read-count specimens.

### Impact by specimen read count

The CER framework’s impact scales with specimen size, consistent with its statistical design:

| Specimen Reads | Specimens | Split Prevention | Summarize Filtering | Variant Reduction |
|----------------|-----------|------------------|---------------------|-------------------|
| < 50           | 101       | -9               | +5                  | 3.1%              |
| 50–99          | 68        | -11              | +4                  | 5.6%              |
| 100–199        | 99        | -33              | +2                  | 14.9%             |
| 200–499        | 225       | -129             | -174                | 39.5%             |
| 500–999        | 344       | -141             | -708                | 53.0%             |

**Low-read specimens (< 100 reads):** CER has minimal effect. The minimum M for significance at 1.5% error requires 30–50% of reads at small N (e.g., M=5 at N=10, M=6 at N=20), so most variants that survive the existing frequency/count filters are already significant. Split prevention accounts for nearly all reduction; summarize filtering actually retains slightly *more* variants than baseline because fewer, larger input clusters merge differently during summarization.

**Mid-range specimens (100–499 reads):** Split prevention and summarize filtering both contribute. The minimum M threshold drops to 5–14% of reads, catching more marginal variants.

**High-read specimens (500+ reads):** Summarize filtering dominates (83% of reduction in this bin). The minimum M threshold drops to ~3% of reads, so many small secondary clusters (RiC 5-12) are still split off by speconsense but then correctly identified as statistically insignificant during summarization.

At low read counts, split prevention can occasionally cause parent clusters to remain above size filtering thresholds, preserving variants that the baseline would have lost after splitting. This explains the small positive summarize filtering values in the low-read bins — a net gain of variants through this indirect mechanism.

### Quality metrics

**Read identity:** Mean RID improved (99.3% → 99.4%). The quality report flagged 124 low-RID sequences in baseline vs. 67 under CER — the eliminated sequences were disproportionately low-quality small variants. Merged sequences with low-RID components dropped from 8 to 1.

## IUPAC ambiguities:

| Level                        | Baseline | CER 0.015 | Delta         |
|------------------------------|----------|-----------|---------------|
| Raw clusters (pre-summarize) | 928      | 1,575     | +647 (+70%)   |
| Summary output               | 1,615    | 1,397     | -218 (-13.5%) |

The two levels tell different stories. Raw cluster ambiguities increase because split prevention absorbs variant reads into parent clusters, encoding the variation as IUPAC codes. But after summarization — where the baseline’s many small variants would have been merged back together, also introducing ambiguities — the net effect is a 13.5% *decrease* in total ambiguities. The ambiguity tradeoff is concentrated in the 500+ read bin (-259), where summarize filtering prevents low-support variants from introducing spurious IUPAC codes during merging.

**High-error positions:** Flagged sequences dropped from 579 to 479 (-17%). Surviving flagged sequences tend to have larger RiC, representing genuine biological complexity rather than artifact.

**Primary variant stability:** 93% of primary (v1) sequences are identical between baseline and CER. Of 80 that changed, 50 gained ambiguities (from absorbed variant reads) and 18 lost them. The largest changes occur in specimens with many secondary variants.

## Edge cases observed

**Small specimens ( $N < 50$ ):** Of 19 multi-variant specimens in this range, CER only affected 3. The framework is appropriately conservative at low  $N$ , deferring to the existing frequency/count filters.

**Specimens gaining raw clusters:** 5 specimens had more raw clusters under CER than baseline. Preventing one split can change the size distribution enough to cause a different split to become significant, or alter the clustering graph topology — a plausible indirect effect.

**Correlated multi-position variation:** One cluster (ONT01.19, size=11) accumulated multiple IUPAC ambiguities after CER prevented splitting reads that differed at correlated positions. The current framework evaluates positions independently ( $K=1$ ), so correlated variation is not modeled. This is acknowledged in the companion paper as out of scope for this version — multi-position variants are at least as significant as single-position variants, so the  $K=1$  analysis is conservative.

## 9. Open questions for reviewers

1. ~~Is 2% the right default assumed error rate?~~ Resolved: default set to 1.5%, slightly below the typical ONT R10 per-position substitution rate of ~2%. This provides a margin that avoids filtering variants near the boundary while still catching clear artifacts. Validated on ONT98: 42% variant reduction with zero specimen loss at 1.5%.
2. **Should the default alpha be 1e-5 or something else?** The paper suggests 1e-5 as a “run-corrected” level (accounting for ~100 specimens per run). A per-specimen alpha of 0.05 would be more permissive.
3. ~~Should CER filtering in speconsense summarize default to enabled?~~ Resolved: yes, defaults to 0.015 to match the core tool. With primary-variant protection, the risk of surprising behavior is minimal — only secondary variants with weak statistical support are affected. Use `--no-cer-filter` to disable.
4. **Is the cer annotation useful for your validation workflows?** Would you use it for triage? Should it appear in additional presets (e.g., `default`)?
5. ~~Edge case: MCL-only clusters.~~ Resolved: all clusters now receive `cer` annotation computed from final cluster size, so MCL-only and phasing-split clusters are treated uniformly.

6. ~~Should CER filtering protect primary variants?~~ Resolved: CER filtering in speconsense summarize now operates after HAC grouping and protects the primary (largest) variant in each group. Single-cluster specimens are never eliminated by CER filtering. This was validated on the ONT98 dataset — zero specimens lost with the updated filtering.

## Appendix: Reference values

Minimum M for significance at default settings (uniform model, L=700, alpha=1e-5, p=1.5%)

| N    | Min M | % of N |
|------|-------|--------|
| 10   | 5     | 50.0%  |
| 15   | 5     | 33.3%  |
| 20   | 6     | 30.0%  |
| 30   | 6     | 20.0%  |
| 50   | 7     | 14.0%  |
| 100  | 9     | 9.0%   |
| 200  | 11    | 5.5%   |
| 500  | 16    | 3.2%   |
| 1000 | 23    | 2.3%   |

At low N, the minimum M is a large fraction of reads — CER is conservative and defers to frequency/count filters. At high N, the threshold drops to a few percent, catching weak variants that the frequency filter cannot.

Minimum M at other operating points (uniform model, L=700)

From the companion paper, Table 2:

| N    | p=1%,<br>alpha=0.05 | p=1%,<br>alpha=1e-5 | p=1.5%,<br>alpha=1e-5 | p=2%,<br>alpha=1e-5 | p=3%,<br>alpha=1e-5 |
|------|---------------------|---------------------|-----------------------|---------------------|---------------------|
| 100  | 5                   | 8                   | 9                     | 10                  | 11                  |
| 500  | 9                   | 14                  | 16                    | 18                  | 22                  |
| 1000 | 13                  | 19                  | 23                    | 26                  | 33                  |

Critical error rate p\* at common operating points (uniform model, L=700)

| N    | M   | alpha | p*     |
|------|-----|-------|--------|
| 1000 | 100 | 0.05  | 20.25% |
| 1000 | 100 | 1e-5  | 16.59% |
| 100  | 10  | 0.05  | 6.55%  |
| 100  | 10  | 1e-5  | 2.50%  |